# Quiz — 1.3.1 Compression, Encryption and Hashing

Name: ____ Date: _ ***Mark:*** ___ / 25

*OCR H446 · Component 01 · 1.3.1*

## Section A — Multiple choice (8 marks)

**1.** Which statement about lossy compression is correct? A. The original file can always be perfectly reconstructed B. Data is permanently removed and cannot be recovered C. It is only used for text files D. It never reduces file size

Your answer: ☐

**2.** Run Length Encoding works best on data that contains: A. Random values with no repetition B. Long runs of repeated identical values C. Encrypted ciphertext D. A single unique value per pixel

Your answer: ☐

**3.** A disadvantage of RLE is that it can: A. Make files larger when there are few repeated runs B. Permanently lose data C. Only compress audio D. Require a public and private key

Your answer: ☐

**4.** In dictionary-based compression, what must be supplied so the file can be decompressed? A. The encryption key B. A hash digest C. The dictionary (index of patterns) D. The damping factor

Your answer: ☐

**5.** In symmetric encryption: A. A public key encrypts and a private key decrypts B. The same key is used to encrypt and decrypt C. No key is needed D. The key is never shared but is also never used

Your answer: ☐

**6.** Which is the main weakness of symmetric encryption that asymmetric encryption solves? A. It is too slow to be practical B. The secure distribution of the shared key C. It produces a fixed-length output D. It cannot encrypt text

Your answer: ☐

**7.** A digital signature is created by: A. Hashing the message with no key B. Signing with the sender's private key (verified with the public key) C. Encrypting with the recipient's public key only D. Using RLE on the message

Your answer: ☐

**8.** Which property means a hash function always produces the same digest for the same input? A. One-way B. Collision-resistant C. Deterministic D. Reversible

Your answer: ☐

## Section B — Short answer (12 marks)

**9.** Define lossless compression and give one suitable use. **[2]**

**10.** Using RLE, encode the following pixel run. Write your encoded output as value+count pairs. **[2]**

`W W W W W B B W W W W W W`

**11.** The following data was encoded with RLE as `5A 3B 1C 4A`. Decode it back to the original sequence. **[2]**

**12.** Explain why asymmetric encryption is described as more secure for exchanging data with a stranger than symmetric encryption. **[2]**

**13.** A website stores user passwords by hashing them. Explain two reasons why hashing (rather than encryption) is used. **[2]**

**14.** State two properties a good hash function should have. **[2]**

## Section C — Exam-style question (5 marks)

**15.** A music streaming service sends audio files to users' devices and stores users' login details on its server.

Describe how compression, encryption and hashing could each be used by this service. Refer to the type of compression suitable for audio, the type of encryption used during data transfer, and how hashing protects stored login details. **[5]**

## Answer key

**Section A (8 marks, 1 each)**

1. **B** — Lossy permanently discards data; the original cannot be exactly recovered.
2. **B** — RLE replaces runs of repeated values, so long runs compress well.
3. **A** — With few/short runs, the value+count pairs can exceed the original size.
4. **C** — The dictionary mapping patterns to references must accompany the file to decode it.

5. **B** — Symmetric uses one shared key for both encryption and decryption.
6. **B** — Symmetric's key-distribution problem is solved by the openly shared public key.
7. **B** — Sign with private key, verify with public key (proves origin/integrity).
8. **C** — Deterministic: same input always yields the same digest.

**Section B (12 marks)**

9. **[2]** 1 mark: no data is permanently removed / original is exactly recoverable. 1 mark: any valid use, e.g. text/program code/ZIP/PNG.

10. **[2]** Encoded output: `5W 2B 6W` (or `W5 B2 W6`). 1 mark correct values W,B,W in order; 1 mark correct counts 5,2,6.

11. **[2]** Decoded: `AAAAA BBB C AAAA` i.e. `AAAAABBBCAAAA`. 1 mark for correct letters in order; 1 mark for correct counts (5,3,1,4).

12. **[2]** 1 mark: the public key can be shared openly so no secret key needs to be exchanged in advance. 1 mark: only the matching private key (kept secret) can decrypt, so an intercepted public key/ciphertext is useless — avoids the symmetric key-distribution problem.

13. **[2]** Any two for 1 mark each: hashing is one-way so the original password cannot be retrieved even if the database is stolen; the system only compares hashes at login so it never needs to store/recover the plaintext; encryption is reversible with a key (a risk if the key is compromised).

14. **[2]** Any two for 1 mark each: one-way/not reversible; deterministic; low/few collisions; fast to compute; produces fixed-length output.

**Section C (5 marks — award up to 5 from)**

15. 
    - Compression reduces audio file size so it transfers faster and uses less bandwidth/storage. **[1]**
    - Lossy compression (e.g. MP3) is suitable for audio because slight quality loss is acceptable and gives a much smaller file. **[1]**
    - Encryption scrambles data in transit so an interceptor cannot read it; (a)symmetric encryption is used — HTTPS uses asymmetric to exchange a symmetric session key, then symmetric for the bulk audio stream. **[1]**
    - Hashing the password means only the hash is stored, not the plaintext. **[1]**
    - At login the entered password is hashed and compared; because hashing is one-way a stolen database does not reveal the actual passwords. **[1]**

**Total: 8 + 12 + 5 = 25 marks**